CU24250181

BTech CSE 3rd Year | Section B

Roll No. :- 11

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Experiment No. 5

Feature Extraction and Image Analysis using SIFT and HOG Descriptors

Aim

To implement SIFT and HOG for feature extraction and analyze their use in image representation, matching, object recognition, and computer vision applications.

Input image used: image.jpg
All output images in this report were generated locally.

Step 1: Import required libraries

Load OpenCV, NumPy, Matplotlib, SIFT, HOG, and image enhancement tools.

Code

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from pathlib import Path
from skimage.feature import hog
from skimage import exposure
```

Step 2: Load and preprocess the image

Read the local image, convert it to RGB for display, and create a grayscale image.

Code

```
image_path = Path(__file__).parent / "image.jpg"
image = cv2.imread(str(image_path))
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
print("Image loaded successfully.")
```

Output: Original input image



Step 3: Convert image to grayscale

Resize the grayscale image to a consistent size for HOG analysis.

Code

```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
gray_resized = cv2.resize(gray, (128, 128))
```

Output: Grayscale image



Step 4: Detect SIFT keypoints and descriptors

SIFT identifies distinctive local points and creates a 128-dimensional descriptor for each point.

Code

```
sift = cv2.SIFT_create()
keypoints, descriptors = sift.detectAndCompute(gray, None)
keypoints_image = cv2.drawKeypoints(
    image_rgb, keypoints, None,
    flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS
)
print("Number of keypoints:", len(keypoints))
print("Descriptor shape:", descriptors.shape)
```

Output

```
Number of keypoints: 17518
Descriptor shape: (17518, 128)
```

Output: SIFT keypoints



Step 5: Extract HOG features

HOG represents local edge directions and object shape using normalized gradient histograms.

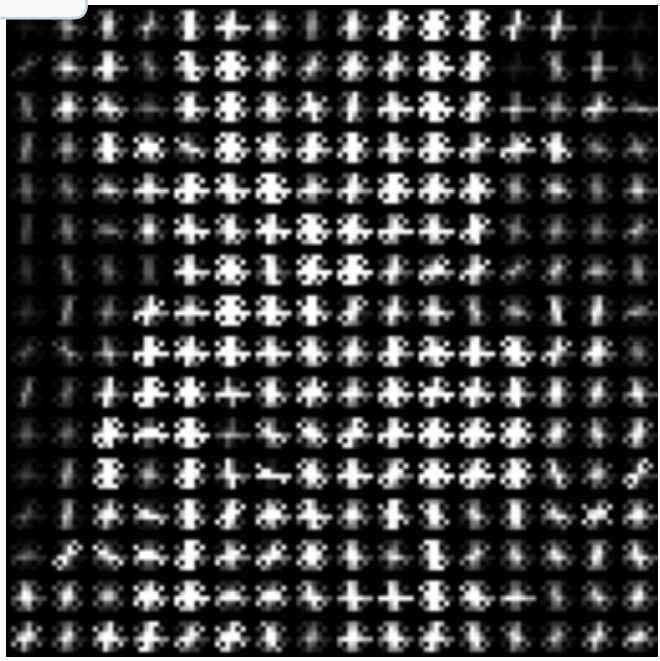
Code

```
gray_resized = cv2.resize(gray, (128, 128))
features, hog_image = hog(
    gray_resized, orientations=9,
    pixels_per_cell=(8, 8), cells_per_block=(2, 2),
    visualize=True
)
hog_image_rescaled = exposure.rescale_intensity(
    hog_image, in_range=(0, 10)
)
print("HOG feature vector length:", len(features))
```

Output

```
HOG feature vector length: 8100
Orientations: 9
Pixels per cell: (8, 8)
Cells per block: (2, 2)
```

Output: HOG visualization



Step 6: Match SIFT features

Compare the original image with a rotated and resized copy using the ratio test.

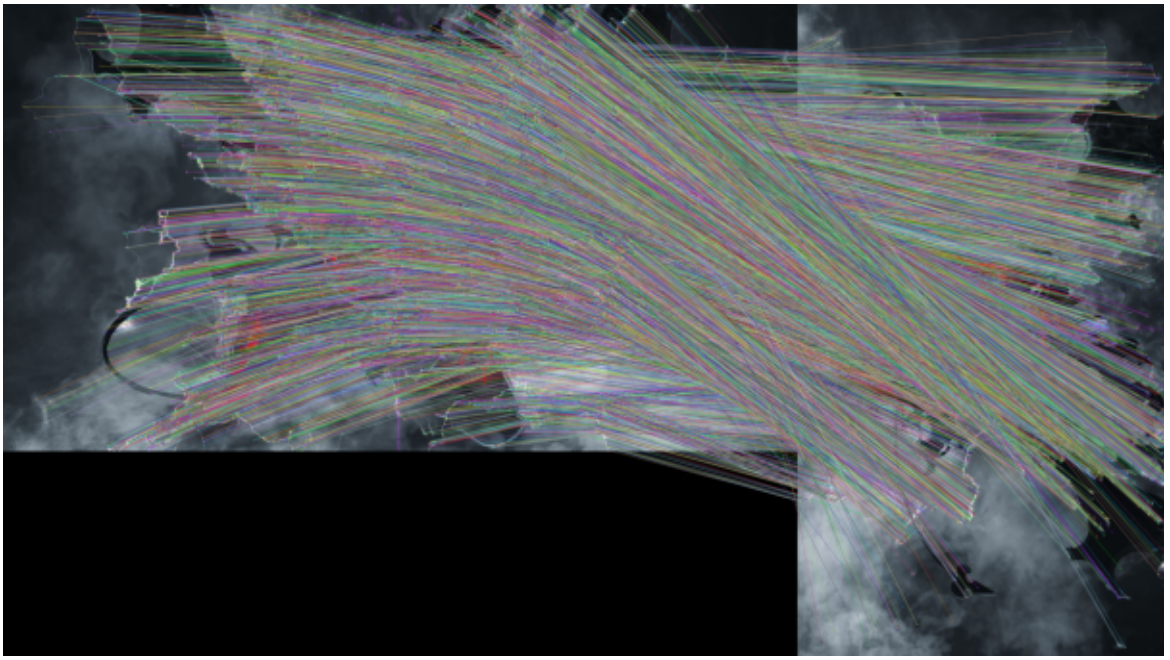
Code

```
transformed = cv2.rotate(
    cv2.resize(image, None, fx=0.82, fy=0.82),
    cv2.ROTATE_90_CLOCKWISE
)
keypoints2, descriptors2 = sift.detectAndCompute(
    cv2.cvtColor(transformed, cv2.COLOR_BGR2GRAY), None
)
matches = cv2.BFMatcher().knnMatch(descriptors, descriptors2, k=2)
good_matches = [m for m, n in matches if m.distance < 0.75 * n.distance]
print("Total matches:", len(matches))
print("Good matches:", len(good_matches))
```

Output

```
Total matches: 17518
Good matches: 10194
```

Output: SIFT feature matching



Experiment output and observations

SIFT keypoints detected: 17518
SIFT descriptor shape: (17518, 128)
HOG feature vector length: 8100
Good SIFT matches with transformed image: 10194

1. SIFT detects distinctive local keypoints and generates descriptors for feature matching.
2. SIFT is robust to image scale and rotation changes.
3. HOG captures edge direction and shape information.
4. HOG is useful for object detection, especially when object shape is important.
5. The number of detected features depends on image content and parameter settings.

Conclusion: SIFT is preferable for local feature matching, while HOG is a compact shape descriptor for detection and classification pipelines.

Answers to questions

1. Feature extraction converts raw pixels into meaningful numerical representations such as edges, corners, shapes, and textures.

2. SIFT constructs a scale space, detects stable keypoints, assigns orientations, and creates descriptors around the keypoints.

3. A keypoint is a distinctive image location. A descriptor is the numerical representation of the local region around it.

4. HOG divides an image into cells, calculates gradient orientation histograms, groups cells into blocks, and normalizes them.

5. SIFT is robust to scale and rotation and is suited to image matching. HOG describes shape and edges and is suited to object detection.

6. SIFT achieves scale invariance by detecting features across multiple scales and rotation invariance by assigning a dominant orientation.

7. HOG is used in pedestrian detection, vehicle detection, and general shape-based object detection.

8. Feature extraction reduces data complexity and highlights patterns that classifiers and detectors can use efficiently.

9. Handcrafted descriptors are interpretable and can work with limited data, while deep learning learns richer features but usually needs more data and computation.

10. SIFT descriptors support image matching and recognition; HOG descriptors support shape-based object detection.